1 Worksheet

Question 1.1. Let $f : E \to [0, \infty]$ satisfy $\int_E f < \infty$. Let $A_n \subseteq A_{n-1} \subseteq E$ be a decreasing sequence of sets such that $m(A_n) \to 0$ as $n \to \infty$. Show that

$$\lim_{n\to\infty} \int_E f\mathbb{1}_{A_n} = 0$$

Proof. Let $\varepsilon > 0$ and let $\varphi : E \to [0, \infty]$ be a simple function satisfying $\varphi \leq f$ and $(\int_E f) - \varepsilon < \int_E \varphi$. We then have the following chain of inequalities for any n:

$$0 \leq \int_E (f - \varphi)\mathbb{1}_{A_n} \leq \int_E (f - \varphi)\mathbb{1}_{A_n} + \int_E (f - \varphi)\mathbb{1}_{A_n^c} = \int_E (f - \varphi) < \varepsilon$$

Now we write $\varphi = \sum_{j=1}^k \alpha_j \mathbb{1}_{E_j}$ for some $E_j \subseteq E$ and get

$$\int_E \varphi\mathbb{1}_{A_n} = \int_E \sum_{j=1}^k \alpha_j \mathbb{1}_{E_j \cap A_n} \leq \sum_{j=1}^k \alpha_j m(A_n)$$

which approaches 0 as $n \to \infty$. We then have

$$\int_E f\mathbb{1}_{A_n} < \int_E \varphi\mathbb{1}_{A_n} + \varepsilon \Longrightarrow 0 \leq \limsup_{n\to\infty} \int_E f\mathbb{1}_{A_n} \leq \lim_{n\to\infty} \int_E \varphi\mathbb{1}_{A_n} + \varepsilon = \varepsilon$$

Since the above holds for all $\varepsilon > 0$, the limsup, hence the limit itself, of $\int_E f\mathbb{1}_{A_n}$ as $n \to \infty$ is 0. □

Question 1.2. Let E be Lebesgue measurable with $m(E) > 0$. Show that there exists $p \in E$ so that $m(E \cap \{x : |x - p| < \varepsilon\}) > 0$ for all $\varepsilon > 0$. (Problem suggested by a student.)

Proof. Let $\varepsilon > 0$ and consider the sets $E_n := (n\varepsilon, (n + 1)\varepsilon) \cap E$ and note that the union of all E_n for $n \in \mathbb{Z}$ is $E \setminus \{n\varepsilon : n \in \mathbb{Z}\}$ which has the same measure as E. We then have

$$m(E) = \sum_{n\in\mathbb{Z}} m(E_n)$$

and since $m(E) > 0$, we cannot have $m(E_n) = 0$ for all n. Thus for some $n \in \mathbb{Z}$, $m(E_n) > 0$, so we can take any $p \in E_n = (n\varepsilon, (n+1)\varepsilon) \cap E$ and we will have $(p - \varepsilon, p + \varepsilon) \supseteq (n\varepsilon, (n+1)\varepsilon)$, hence $m(E \cap (p - \varepsilon, p + \varepsilon)) \geq m(E_n) > 0$ as required. □

Question 1.3. Let $f_n : \mathbb{R} \to [0, \infty]$ be a sequence of Lebesgue measurable functions that converge pointwise to some f. Assume that

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f < \infty$$

Show that

$$\lim_{n\to\infty} \int_{\mathbb{R}} \mathbb{1}_E f_n = \int_{\mathbb{R}} \mathbb{1}_E f$$

for every measurable E. Show that this may fail if $\lim_{n\to\infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f = \infty$.

Proof. Note that if we show that $\int_{\mathbb{R}} |f_n - f| \to 0$, then we will have

$$0 \leq \left| \int_{\mathbb{R}} \mathbb{1}_E f_n - \int_{\mathbb{R}} \mathbb{1}_E f \right| \leq \int_{\mathbb{R}} \mathbb{1}_E |f_n - f| \leq \int_{\mathbb{R}} |f_n - f| \to 0$$

and hence the desired limit identity holds. For $\int_{\mathbb{R}} |f_n - f| \to 0$, we note that $f + f_n - |f_n - f|$ is either f or f_n for each x, hence nonnegative, and its liminf is $2f$. We thus have by Fatou's lemma

$$\begin{aligned}
2\int_{\mathbb{R}} f &= \int_{\mathbb{R}} \liminf_{n\to\infty}(f + f_n - |f_n - f|) \\
&\leq \liminf_{n\to\infty} \int_{\mathbb{R}} (f + f_n - |f_n - f|) \\
&= \int_{\mathbb{R}} f + \liminf_{n\to\infty} \int_{\mathbb{R}} f_n - \limsup_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \\
&= 2\int_{\mathbb{R}} f - \limsup_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \\
\implies 0 \leq \lim_{n\to\infty} \int_{\mathbb{R}} |f_n - f| &\leq \limsup_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \leq 0
\end{aligned}$$

Thus we see that $\lim_{n\to\infty} \int_{\mathbb{R}} |f_n - f| = 0$, which completes the proof. □

Question 1.4. Let $f_n : E \to [0, \infty]$ be a decreasing sequence of Lebesgue measurable functions such that $\int_E f_1 < \infty$. Let f be the pointwise limit. Show that

$$\lim_{n\to\infty} \int_E f_n = \int_E f$$

Proof. Just apply DCT with $g = f_1$. □

Question 1.5. If $f : \mathbb{R} \to \overline{\mathbb{R}}$ is integrable, show that the function

$$F(x) := \int_{\mathbb{R}} \mathbb{1}_{(-\infty,x)} f$$

is continuous.

Proof. Let $x \in \mathbb{R}, \varepsilon > 0$, and consider the measurable sets $A_n = [x, x + 1/n)$, which is a decreasing sequence with $m(A_n) \to 0$. Note that $\int_{\mathbb{R}} |f| < \infty$ by integrability of f, so we can apply question 1 to get

$$\left| F\left(x + \frac{1}{n}\right) - F(x) \right| = \left| \int_{\mathbb{R}} \mathbb{1}_{[x,x+1/n)} f \right| \leq \int_{\mathbb{R}} \mathbb{1}_{[x,x+1/n)} |f| \to 0$$

we can thus take $N > 0$ for which $\left|F\left(x + \frac{1}{N}\right) - F(x)\right| < \varepsilon$. It then follows that for all $h \in (0, 1/N)$, we have

$$|F(x+h) - F(x)| \leq \int_{\mathbb{R}} \mathbb{1}_{[x,x+h)} |f| \leq \int_{\mathbb{R}} \mathbb{1}_{[x,x+1/N)} |f| < \varepsilon$$

We can do the same for $A_n = (x - 1/n, x]$ and combine with the above to get a $\delta > 0$ for which $|F(y) - F(x)| < \varepsilon$ for all $y \in (x - \delta, x + \delta)$. Therefore, F is continuous. □

2 Royden Chapter 4

Question 2.1 (Royden 4.20)**.** Let $\{f_n\}$ be a sequence of nonnegative measurable functions that converges to f pointwise on E. Let $M \geq 0$ be such that

$$\int_E f_n \leq M \quad \text{for all } n.$$

Show that $\int_E f \leq M$. Verify that this property is equivalent to the statement of Fatou's Lemma.

Proof. □

Question 2.2 (Royden 4.21)**.** Let the function f be nonnegative and integrable over E and $\epsilon > 0$. Show there is a simple function η on E that has finite support, $0 \leq \eta \leq f$ on E, and

$$\int_E |f - \eta| < \epsilon.$$

If E is a closed, bounded interval, show there is a step function h on E that has finite support and

$$\int_E |f - h| < \epsilon.$$

Proof. □

Question 2.3 (Royden 4.25)**.** Let $\{f_n\}$ be a sequence of nonnegative measurable functions on E that converges pointwise on E to f. Suppose $f_n \leq f$ on E for each n. Show that

$$\lim_{n\to\infty} \int_E f_n = \int_E f.$$

Proof. □

Question 2.4 (Royden 4.28)**.** Let f be integrable over E and C a measurable subset of E. Show that

$$\int_C f = \int_E f \cdot \chi_C.$$

Proof. □